Practice Problem: Procrastination and Present Bias

Sophisticated, Naive and Partially Naive Students

Professor Daeyoung Jeong

Yonsei University

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Setup

Quasi-hyperbolic discounting

$$U^t(c_t, c_{t+1}, \ldots, c_T) = u(c_t) + \beta \sum_{\tau=1}^{T-t} \delta^\tau u(c_{t+\tau})$$

- Linear per-period utility $u(c) = c$, long-run discount factor $\delta = 1$, true present bias $\beta = 0.6$.
- A student must complete a difficult assignment. Completing it costs **effort** (a negative payoff) in the period it is done, and yields a **reward of 25** in the following period.
- The effort cost is 12 whenever the task is done.

Types of students

Sophisticated: knows her true β and correctly predicts her future selves. **Naive:** believes her future selves will have no present bias ($\hat{\beta} = 1$). **Partially naive:** believes $\hat{\beta} \in (\beta, 1)$.

Questions

The assignment can be done at $t = 1$ (cost at $t = 1$, reward at $t = 2$) or at $t = 2$ (cost at $t = 2$, reward at $t = 3$). The student is sophisticated.

(a) From the perspective of $t = 0$, calculate U^0(Do at $t = 1$) and U^0(Do at $t = 2$). Which option does the student prefer?

(b) Move to $t = 1$, where the student actually decides. Calculate U^1(Do at $t = 1$) and U^1(Do at $t = 2$). Does the student have the same preference as at $t = 0$? Explain the behavioral mechanism.

(c) Would your answers to (a) and (b) change if the student were **naive** instead of sophisticated? Explain why or why not.

(d) At $t = 0$, what does each type of student (sophisticated, naive, and partially naive with $\hat{\beta} = 0.9$) **predict** she will choose at $t = 1$? Compare each prediction with her preference in (a) and with her actual choice in (b).

What Each Question Really Asks

	The question in the student's own words	What it captures
(a)	"If I had to **commit now** to doing the work at $t = 1$ or $t = 2$, which would I choose?"	The **plan**: today's preferences over two future options.
(b)	"Now that $t = 1$ has arrived, should I do it **now** or **later**?"	The **actual choice**, made with $t = 1$ preferences.
(c)	"Does it matter whether I am **aware** of my present bias?"	Whether beliefs about future selves change any of these **utilities**.
(d)	"When $t = 1$ arrives and I decide then, **what will I choose**?"	The **prediction**: a forecast of a future self's choice, using *beliefs* ($\hat{\beta}$) about that self.

In short

(a) asks *what do I want my future self to do?* (d) asks *what will my future self actually do?* (b) shows what she *does* do.